# COMPOSITE MATERIAL DATASHEETS GFRP 7781/8552-8HS



## MATERIAL IDENTIFICATION

| | |
|---|---|
| Short informal name | 7781/8552-8HS |
| Commercial name | Hexcel Hexply 8552/37%/7781 |
| Specifications | Airbus: AIMS05-02-003 C (ABS5009 F40) |
| Description | Glass Fibre Reinforced Polymer (GFRP) prepreg fabric produced by hand-lay-up |
| Version | 1.0 (30/11/2022) |

## REFERENCES

- QTR-S-M-268 issue 2 "Qualification Test Report for Hexcel 7781-8552S-8HS and 7781-8552-8HS" Airbus Technical Report
- X029RP1439435 issue 11.3 "Summary of Design Values for Composite Materials" Airbus Technical Report
- Airbus DS software COMALL v.0.23.

## PHYSICAL PROPERTIES

| | | |
|---|---|---|
| Cured Ply Thickness (*CPT*) | 0.239 | mm |
| Density | 1880 | $kg/m^3$ |
| Tg-onset (wet) | >130 | ºC |
| Max. Operational Limit (*MOL*) | 120 | ºC |

| | Range | -55…20ºC | 20…70ºC | 70…120ºC |
|---|---|---|---|---|
| Coefficients of Thermal Expansion CTE (1/ºC) | $\alpha_1$ | $13.9\cdot10^{-6}$ | $13.9\cdot10^{-6}$ | $13.9\cdot10^{-6}$ |
| | $\alpha_2$ | $13.9\cdot10^{-6}$ | $13.9\cdot10^{-6}$ | $13.9\cdot10^{-6}$ |
| | $\alpha_3$ | $34.0\cdot10^{-6}$ | $34.0\cdot10^{-6}$ | $34.0\cdot10^{-6}$ |

## ELASTIC MODULI

| Property (Units) | $E_1$ (MPa) | $E_2$ (MPa) | $G_{12}$ (MPa) | $\nu_{12}$ - | $K_B$ - |
|---|---|---|---|---|---|
| Values | 25700 | 25700 | 4300 | 0.050 | 0.865 |

| Property (Units) | $E_3$ (MPa) | $G_{13}$ (MPa) | $G_{23}$ (MPa) | $\nu_{13}$ - | $\nu_{23}$ - |
|---|---|---|---|---|---|
| Values | (1) | (1) | (1) | (1) | (1) |

*Remarks:*

- Values in tables (as well as CPT and MOL above) are in line with values proposed by Airbus Commercial, for commonality and convenience. Dedicated derivation from qualification tests reports was not leading to relevant advantages, and values marked (1) cannot be accurately derived.
- The $K_B$ is the buckling correction factor for moduli to apply to $E_1$, $E_2$ and $G_{12}$ in buckling stability analyses.

## PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

| Property | Mean Values (MPa) | | | | | B-basis | *E-KDF* | | | B-basis Values (MPa) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | -55ºC Dry | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet | $K_B$ | 70ºC Wet | 90ºC Wet | 120ºC Wet | -55ºC Dry | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet |
| $F_{1t}$ | 547.4 | 414.2 | 285.0 | 272.3 | - | 0.900 | 0.688 | 0.657 | - | 492.7 | 372.8 | 256.5 | 245.1 | - |
| $F_{1c}$ | 887.7 | 659.7 | 434.1 | 405.9 | - | 0.900 | 0.658 | 0.615 | - | 799.0 | 593.7 | 390.7 | 365.3 | - |
| $F_{2t}$ | 547.4 | 414.2 | 285.0 | 272.3 | - | 0.900 | 0.688 | 0.657 | - | 492.7 | 372.8 | 256.5 | 245.1 | - |
| $F_{2c}$ | 887.7 | 659.7 | 434.1 | 405.9 | - | 0.900 | 0.658 | 0.615 | - | 799.0 | 593.7 | 390.7 | 365.3 | - |
| $F_{12}$ | 127.2 | 110.2 | 79.2 | 72.0 | 57.8 | 0.599 | 0.718 | 0.653 | 0.525 | 76.2 | 66.0 | 47.4 | 43.1 | 34.6 |

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

| Property | Mean Values (με) | | | | | B-basis | *E-KDF* | | | B-basis Values (με) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | -55ºC Dry | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet | $K_B$ | 70ºC Wet | 90ºC Wet | 120ºC Wet | -55ºC Dry | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet |
| $\varepsilon_{tall}$ | 15900 | 15900 | 11100 | 10200 | - | 0.853 | 0.698 | 0.642 | - | 13600 | 13600 | 9500 | 8700 | - |
| $\varepsilon_{call}$ | 25700 | 25700 | 16900 | 15800 | - | 0.900 | 0.658 | 0.615 | - | 23100 | 23100 | 15200 | 14200 | - |

## INTERLAMINAR PLY STRENGTHS

| Property (Units) | Mean Values | | | | | B-basis | *E-KDF* | | | B-basis Values | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | -55ºC Dry | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet | $K_B$ | 70ºC Wet | 90ºC Wet | 120ºC Wet | -55ºC Dry | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet |
| $F_{13}$ (MPa) | 100.9 | 81.6 | 53.8 | 46.5 | 36.9 | 0.729 | 0.659 | 0.570 | 0.452 | 73.6 | 59.5 | 39.2 | 33.9 | 26.9 |
| $F_{3t}$ (MPa) | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| $G_{IC}$ (N/mm) | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| $G_{IIC}$ (N/mm) | - | - | - | - | - | - | - | - | - | - | - | - | - | - |

## OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

| Property | Mean Values (MPa) | | | | B-basis | *E-KDF* | | | B-basis Values (MPa) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet | $K_B$ | 70ºC Wet | 90ºC Wet | 120ºC Wet | RT Dry | 70ºC Wet | 90ºC Wet | 120ºC Wet |
| $OHT_{QI}$ | 177.0 | 109.1 | 108.7 | - | 0.900 | 0.617 | 0.614 | - | 159.3 | 98.2 | 97.9 | - |
| $OHC_{QI}$ | 272.0 | - | 167.2 | - | 0.900 | - | 0.615 | - | 244.8 | - | 150.5 | - |
| $FHT_{QI}$ | 192.8 | 131.8 | 128.2 | - | 0.900 | 0.684 | 0.665 | - | 173.5 | 118.6 | 115.4 | - |
| $FHC_{QI}$ | 463.7 | 307.7 | 269.6 | - | 0.900 | 0.664 | 0.581 | - | 417.4 | 276.9 | 242.6 | - |
| $F_{BR\text{-}QI}$ | 557.0 | 349.7 | 204.7 | 204.7 | 0.833 | 0.628 | 0.367 | 0.367 | 463.8 | 291.1 | 170.4 | 170.4 |
| $F_{PO}$ | - | - | - | - | - | - | - | - | - | - | - | - |

## COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression (as well as approximation for dent depth to impact energy).

| Property | B-basis Value for all conditions |
|---|---|
| *TAI* (με) | SMM §1421 → $\varepsilon_{tall}$ / 3 = 9500 / 3 ≈ **3100** |

In compression, the applicable approximation is to use the Open-Hole analysis, i.e. failure when hole tangent strain is equal to allowable fibre strain (*) as a replacement of the CAI strength.

(*) No relaxations as the “characteristic distance” approach (popular in literature for open-hole failure predictions) shall be applied to the hole edge strain in this particular problem.

(**) CAI strengths obtained according to abovementioned approach is shown conservative according to trend shown by CAI qualification tests on the quasi-isotropic laminate of this glass fibre laminate (however the CAI strength after full penetration of impactor is not available to build a more sophisticated approximation model). On the other hand, typical values of CAI with proposed approximation will be higher than design value for TAI, what is another signal of the wide conservative adopted for tension cases.